Advanced Statistics 2027-1: Syllabus Basis

Working basis for developing the course syllabus and first lecture.

Course Frame

The course will be based primarily on Kevin Murphy's *Probabilistic Machine Learning: Advanced Topics* covering selected material in chapters 1-33. The main intellectual goal is to present machine learning as a coherent extension of probabilistic and statistical modeling, rather than as a collection of ad hoc algorithms. The course should build naturally on the earlier Bayesian/statistics course. Students should see prediction, generation, and discovery as connected modeling tasks with shared probabilistic foundations. The key topics covered include:

Foundations of probabilistic and Bayesian machine learning: probabilistic modeling, graphical models; optimization and approximate inference; Monte Carlo methods

Predictive modeling: generalized linear models; deep neural networks; Bayesian neural networks; Gaussian processes

Generative modeling: variational autoencoders; autoregressive models; generative adversarial networks

Latent-variable and representation learning: latent factor models; state-space models; representation learning; interpretable machine learning

Chosen Course Shape

We will use a 13-meeting structure:

- 12 content lectures
- 1 project presentation

Lecture Mapping

Lecture	Unit	Material	Murphy book chapters
1	Fundamentals	Probability + Bayesian statistics	Ch. 1, *Introduction*; Ch. 2, *Probability*; Ch. 3, *Statistics*
2	Fundamentals	Graphical models + information theory	Ch. 4, *Graphical Models*; Ch. 5, *Information Theory*
3	Fundamentals	Optimization	Ch. 6, *Optimization*
4	Inference	Gaussian filtering/smoothing + message passing	Ch. 7, *Inference Algorithms: An Overview*; Ch. 8, *Gaussian Filtering and Smoothing*; Ch. 9, *Message Passing Algorithms*
5	Inference	Variational inference	Ch. 10, *Variational Inference*
6	Inference	Monte Carlo + MCMC	Ch. 11, *Monte Carlo Methods*; Ch. 12, *Markov Chain Monte Carlo*
7	Prediction	GLMs + deep neural networks	Ch. 14, *Predictive Models: An Overview*; Ch. 15, *Generalized Linear Models*; Ch. 16, *Deep Neural Networks*
8	Prediction	Bayesian neural networks + Gaussian processes	Ch. 17, *Bayesian Neural Networks*; Ch. 18, *Gaussian Processes*
9	Generation	Generative modeling overview + VAEs	Ch. 20, *Generative Models: An Overview*; Ch. 21, *Variational Autoencoders*
10	Generation	Autoregressive models + GANs	Ch. 22, *Autoregressive Models*; Ch. 23, *Normalizing Flows*; Ch. 24, *Energy-Based Models*; Ch. 25, *Diffusion Models*; Ch. 26, *Generative Adversarial Networks*
11	Discovery	Latent factor models + state-space models	Ch. 27, *Discovery Methods: An Overview*; Ch. 28, *Latent Factor Models*; Ch. 29, *State-Space Models*; Ch. 30, *Graph Learning*; Ch. 31, *Nonparametric Bayesian Models*

Lecture	Unit	Material	Murphy book chapters
12	Discovery	Representation learning + interpretable ML	Ch. 32, *Representation Learning*; Ch. 33, *Interpretability*
13	Integration	Project presentations / synthesis	—

Course Format

The weekly structure is:

- 2 hours lecture
- 1 hour notebook-based self-work or guided work

Assessment

- AI-based tutoring: 10%
- Notebook submission: 10%
- Project presentation: 30%
- Final exam: 50%